Animal incursions worldwide (2001-2025)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Africa	South Africa	Jul-03	Horse	Africa	Zimbabwe	5 months	Human-mediated	International (shared border)	By land	Imported from Harare to Pietermaritzburg	No		No local transmission	No	Sabeta & Randles, 2005 (1)
Africa	Ethiopia	Sep-03	Dog	Africa	Ethiopia		Natural	In-country	By land	Incursion from nearby towns that caused rabies outbreak in Ethiopian wolves of Bale Mountains National Park	No	Trap-vaccination-release of Ethiopian wolves; controlled by 2004	Outbreak	Yes	Laurenson et al., 2005 (2)
Africa	South Africa	Aug-05	Dog	Africa	Zimbabwe		Natural	International (shared border)	By land	Caused sharp increase in dog rabies cases in Limpopo province	Yes	Mass dog vaccination	No local transmission	No	Zulu et al., 2008 (3); Sabeta et al., 2011 (4); Townsend et al., 2013 (5)
Africa	South Africa	2008	Dog	Africa	South Africa		Natural	In-country	By land	Suspected incursion from Nkomazi that led to outbreak in other parts of Mpumalanga	Yes		Outbreak	No	Phahladira et al., 2012 (6)
Africa	South Africa	2010	Dog	Africa	South Africa			In-country	By land (presumed)	Imported from KwaZulu-Natal and bit local pet dog in Witpoortjie, resulting in outbreak in Gauteng Province	No		Outbreak	No	Sabeta et al., 2013 (7); Weyer et al., 2020 (8); Ngoepe et al., 2022 (9)
Africa	South Africa	2011	Dog	Africa	Lesotho			International (shared border)	By land (presumed)	Suspected incursion into Sisonke district, KwaZulu-Natal	Yes		No local transmission	No	Mollentze et al., 2013 (10)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Africa	South Africa	2012	Dog	Africa	South Africa		Natural	In-country	By land (presumed)	Suspected incursion from North West Province that led to jackal rabies outbreak in uThukela, KwaZulu-Natal	Yes		No local transmission	No	Ngoepe et al., 2022 (9)
Africa	South Africa	2012	Dog	Africa	Lesotho			International (shared border)	By land (presumed)	Rescued stray; secondary transmission to 1 human (owner)	Yes		Secondary transmission	No	Mollentze et al., 2013 (10)
Africa	Chad	2014	Dog	Africa	Chad		Human-mediated	In-country	By land (presumed)	Suspected human-mediated incursion into N'Djamena from surrounding area	Yes		No local transmission	No	Zinsstag et al., 2017 (11)
Africa	Tanzania	Sep-16	Dog	Africa	Tanzania			In-country	By water (presumed)	Suspected incursion from Zanzibar that led to outbreak in Pemba facilitated by lapse in dog mass vaccination	Yes	Mass dog vaccination, PEP (subsidized); outbreak ended in 2018	Outbreak	No	Lushasi et al., 2022 (12)
Africa	Tanzania	Sep-16	Dog	Africa	Tanzania			In-country	By water(presumed)	Suspected incursion from Serengeti District indicated by phylogenetic analysis; led to outbreak in Pemba during lapse in mass dog vaccination; resulted in human rabies deaths	Yes	Mass dog vaccination, PEP (subsidized); outbreak ended in 2018	Outbreak	No	Lushasi et al., 2023 (12), Hayes et al., 2025 (13)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Africa	South Africa	2016	Jackal	Africa	South Africa		Natural	In-country	By land	Suspected incursion from North West Province that led to outbreak in Gauteng Province	Yes	PEP	Outbreak	No	Ngoepe et al., 2022 (9)
Africa	Ethiopia	2019	Ethiopian wolf	Africa	Ethiopia		Natural	In-country	By land	Suspected incursion from Web Valley to Sanetti	No		No local transmission	No	Marino et al., 2024 (14)
Africa	South Africa	Jul-21	Dog	Africa	South Africa			In-country	By land	Suspected incursion from KwaZulu-Natal Province into Nelson Mandela Bay; led to outbreak	No	Mass dog vaccination, increased public awareness	Outbreak	No	Ravensberg et al., 2022 (15)
Africa	Nigeria	2021	Dog	Africa	Nigeria			In-country	By land	Suspected incursions from Bauchi to Plateau state, resulting in repeated outbreaks attributed to neglect of mass dog vaccination	No		Outbreak	No	Williams et al., 2025 (16)
Asia	Iran	Jan-01	Dog	Asia	Nepal	5 days	Human-mediated	International	By air	Vaccinated against rabies after importation	No	PEP (owners)	No local transmission	No	Johnson et al., 2011 (17)
Asia	Indonesia	2001	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Flores to Lembata	No		Outbreak	No	Ward and Wera, 2025 (18)
Asia	Israel	Mar-03	Dog	Asia	Israel		Human-mediated	In-country	By land	Newly adopted by owner during trip to Beer Sheva and brought back to residence	No	Contact tracing, PEP, increased public awareness (tourists)	No local transmission	No	David et al., 2004 (19); David and Yakobson, 2011 (20)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Asia	Indonesia	Aug-03	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Sulawesi to Maluku islands for meat trade	No		No local transmission	No	Townsend et al., 2013 (5)
Asia	Indonesia	2003	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Ambon Island to Seram Island by fishermen or for dog meat trade	No		Outbreak	No	Ward and Wera, 2025 (18)
Asia	Indonesia	2004	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Ambon Island to Buru Island by fishermen or for dog meat trade	No		Outbreak	No	Ward and Wera, 2025 (18)
Asia	Bhutan	May-05	Dog	Asia	India		Natural	International (shared border)	By land	Free-roaming dog that bit cow in Gongza village, resulting in outbreak that spread to other villages accessible to free-roaming dogs through road network	No	Mass dog vaccination, PEP, dog-catching, vaccination of exposed animals, increased public awareness	Outbreak	Yes	Tenzin et al., 2011 (21); Townsend et al., 2013 (5); Rinchen et al., 2020 (22); Rinchen et al., 2020 (23)
Asia	China	Feb-06	Dog	Asia	China			In-country	By land	Suspected incursion from Hengshan city into Yongzhou city; led to secondary transmission to local free-roaming owned dog, which visited neighbouring farm and attacked pigs, causing outbreak with 20 pig deaths out of >50 exposed as	No		Outbreak	Yes	Jiang et al., 2008 (24)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										no control measures were initiated					
Asia	Bhutan	2007	Dog	Asia	India		Natural	International (shared border)	By land	Incursion into Dala subdistrict, Chhukha, Bhutan	No	Culling; controlled by 2008	No local transmission	No	Townsend et al., 2013 (5)
Asia	Indonesia	2008	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Flores to Bali by fishermen with no border controls; bit owners and caused outbreak that spread to other regions with >200 human deaths including owner; delayed detection and response due to poor bite surveillance, non-existent PEP policies, insufficient vaccine stocks and prioritisation of culling	No	Culling, mass dog vaccination	Outbreak	No	Scott-Orr and Putra, 2009 (25); Clifton, 2010 (26); Batan et al., 2014 (27); Putra et al., 2013 (28); Mahardika et al., 2014 (29); Dibia et al., 2015 (30); Cliquet and Wasniewski, 2018 (31); De Jong et al., 2018 (32); Drake, 2020 (33); Rupprecht et al., 2020 (34); Ward and Brookes, 2021 (35)
Asia	Bhutan	2008	Dog	Asia	Bhutan		Natural	In-country	By land	Incursion from south into southwest region	No		No local transmission	No	Rinchen et al., 2020 (22); Rinchen et al., 2020 (23)
Asia	China	Mar-09	Dog	Asia	China		Human-mediated	In-country	By land	Suspected incursion from Sichuan Province; transport of dogs for meat trade enabled outbreak that resulted in >7,000 biting incidents and 20 human deaths	Yes	Culling, mass dog vaccination (limited), PEP (limited)	Outbreak	No	Zhao et al., 2011 (36); Zhang et al., 2014 (37);

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Asia	Indonesia	Feb-10	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Incursion from Sumatra into Nias; led to outbreak that caused >20 human deaths	No	Culling, mass dog vaccination	Outbreak	No	Senior, 2012 (38); Townsend et al., 2013 (5); Rupprecht et al., 2018 (39)
Asia	China	2010	Dog	Asia	China			In-country	By land	Suspected incursion from Hebei Province into Shanxi Province; led to secondary transmission to local owned dog which attacked neighbour's sheep, causing outbreak; spread attributed to high numbers of unvaccinated free-roaming dogs	Yes	Quarantine of exposed animals; sacrifice of associated animals (dog, sheep), mandatory vaccination of pets (owned dogs in the vicinity)	Outbreak	Yes	Zhu et al., 2011 (40)
Asia	China	Dec-10	Dog	Asia	China		Natural	In-country	By land	Free-roaming; suspected incursion from Hebei Province into Shaanxi Province indicated by phylogenetic analysis; detected after secondary transmission to dog, which attacked neighbour's sheep and dog; spread due to unvaccinated free-roaming dogs, lack of border control and meat trade facilitating movement of dogs	Yes	Vaccination of exposed animals (dogs), sacrifice of associated animals (sheep)	Secondary transmission	No	Zhang et al., 2014 (37)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Asia	Indonesia	2010	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Ambon Island to Larat Island by fishermen or for dog meat trade	No		Outbreak	No	Ward and Wera, 2025 (18)
Asia	Russia	Feb-11	Fox	Asia	Mongolia		Natural	International (shared border)	By land, water (frozen)	Suspected incursion through crossing of frozen river; led to outbreak	Yes		Outbreak	No	Adelshin et al., 2012 (41); Adelshin et al., 2015 (42)
Asia	Israel	Oct-11	Dog	Asia	Israel	2 months	Human-mediated	In-country	By land (presumed)	Unvaccinated; brought by owners on camping trip to coast of Sea of Galilee and Northern Israel and stayed outside, was bitten by local rabid stray dog and manifested symptoms after return to Israel	No	Contact tracing, PEP	No local transmission	No	David et al., 2012 (43)
Asia	Philippines	2011	Dog	Asia	Philippines		Human-mediated	In-country	By water (presumed)	Suspected long-distance migration from Central Luzon region to Tablas Island, Romblon; led to outbreak	Yes	Outbreak ended in 2012	Outbreak	No	Tohma et al., 2016 (44)
Asia	Indonesia	2012	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Ambon Island to Kisar Island by fishermen or for dog meat trade	No		Outbreak	No	Ward and Wera, 2025 (18)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Asia	Russia	Feb-12	Fox	Asia	Mongolia		Natural	International (shared border)	By land, water (frozen)	Suspected incursion into through crossing of frozen river	Yes		No local transmission	No	Adelshin et al., 2015 (42)
Asia	South Korea	Apr-13	Dog	Asia	South Korea		Natural	In-country	By land	Suspected incursion from Gangwon Province into Geyonggi Province	Yes		No local transmission	No	Oem et al., 2014 (45)
Asia	Israel	Sep-13	Cow	Asia	Israel	2 weeks	Human-mediated	In-country	By land	Suspected infection by jackal that had been shot in the vicinity and tested positive for rabies; 3 among group of >50 dairy calves transported from dairy farm in Kibbutz Ortal to Jezreel Valley; 2 out of 3 later transported to Kfar Yehoshua	No		No local transmission	No	David et al., 2015 (46)
Asia	China	2014	Red fox	Asia	Russia or Mongolia		Natural	International (shared border)	By land	Suspected incursion that led to secondary transmission to cow	Yes		Secondary transmission	Yes	Liu et al., 2014 (47)
Asia	China	2014	Raccoon dog	Asia	Russia or Mongolia		Natural	International (shared border)	By land	Suspected incursion that led to secondary transmission to goat	Yes		Secondary transmission	Yes	Liu et al., 2014 (47)
Asia	Malaysia	2015	Dog	Asia	Thailand			International (shared border)	By land		Yes		No local transmission	No	Leow et al., 2021 (48)
Asia	Malaysia	2015	Dog	Asia	Thailand			International (shared border)	By land		Yes		No local transmission	No	Leow et al., 2021 (48)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Asia	Bhutan	Oct-16	Dog	Asia	Bhutan		Natural	In-country	By land	Stray, free-roaming; incursion from Meral and Sakteng subdistricts into Rangjung town; bit 3 humans and caused outbreak with >20 infected animals (dogs, cattle, cat); spread facilitated by large stray dog population and lack of public health response, as cases were previously detected in neighboring villages 2-3 months prior	No	PEP	Outbreak	Yes	Tenzin et al., 2017 (49)
Asia	India	Feb-16	Jackal	Asia	India		Natural	In-country	By land	Incursion from West Bengal into Sikkim; bit 2 cows, causing secondary transmission; suspected infection by free-roaming dog while searching for food during winter, as dog vaccination coverage is low due to cultural beliefs that dogs should not be touched	No		Secondary transmission	Yes	Byrnes et al., 2017 (50)
Asia	Malaysia	Jul-17	Dog	Asia	Indonesia		Human-mediated	International (shared border)	By land	Suspected incursion attributed to workers bringing dogs during construction of Pan Borneo Highway; bit 2 children; spread facilitated by minimal rabies control at	Yes	Culling, mass dog vaccination, PEP	Outbreak	No	Rupprecht et al., 2018 (39); Taib et al., 2019 (51); Leow et al., 2021 (48); Ward and Brookes, 2021 (35); Senior, 2012 (38)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										border and interaction of owned and stray dogs, resulting in outbreak					
Asia	Israel	Oct-17	Golden jackal	Asia	Jordan		Natural	International (shared border)	By land	Incursion from Jordan into Jezreel and Hama'ayanot Valleys, Israel; led to outbreak	No	Culling, increased public awareness, ORV (intensified); outbreak ended in 2022	Outbreak	Yes	King et al., 2024 (52)
Asia	Indonesia	2019	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Imported from Bali or Sulawesi to Sumbawa Island to guard corn crops from wild monkeys	No		No local transmission	No	Ward and Brookes, 2021 (35)
Asia	Philippines	2019	Dog	Asia	Philippines		Human-mediated	In-country	By water	Suspected human-mediated incursion from Davao Region to Samal Island indicated by phylogenetic analysis	Yes		Secondary transmission	No	Capin et al., 2023 (53)
Asia	Philippines	2020	Dog	Asia	Philippines		Human-mediated	In-country	By land	Suspected incursion from Davao Del Norte to Davao City indicated by phylogenetic analysis	Yes		Secondary transmission	No	Capin et al., 2023 (53)
Asia	Philippines	Jul-21	Dog	Asia	Philippines		Human-mediated	In-country	By land/water (presumed)	Imported from Central Luzon to Tablas Island, resulted in outbreak; attributed to weak border controls and lapse in annual mass dog vaccination due to COVID-19	Yes	Contact tracing, PEP, increased public awareness, mass dog vaccination	Outbreak	No	Yuson et al., 2024 (54)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										pandemic; resulted in human deaths					
Asia	Philippines	Jun-22	Dog	Asia	Philippines		Human-mediated	In-country	By land/water (presumed)	Imported from Calabarzon region to Tablas Island, resulted in outbreak; attributed to weak border controls and lapse in annual mass dog vaccination due to COVID-19 pandemic; resulted in human deaths	Yes	Contact tracing, PEP, increased public awareness, mass dog vaccination	Outbreak	No	Yuson et al., 2024 (54)
Asia	Philippines		Dog	Asia	Philippines		Human-mediated	In-country	By land/water (presumed)	Imported from Calabarzon region to Marinduque, resulted in outbreak; attributed to weak border controls and lapse in annual mass dog vaccination due to COVID-19 pandemic	Yes	PEP, mass dog vaccination	Outbreak	Yes	Yuson et al., 2024 (54)
Asia	India	2023	Mule	Asia	India		Human-mediated	In-country	By land	Imported from East Sikkim to West Bengal	No	Quarantine, controlled movement of animals	Outbreak	Yes	Aneesh 2023 (55)
Asia	Indonesia	Apr-23	Dog	Asia	Indonesia		Human-mediated	In-country	By water	Owned; imported from Flores into South Timor Tengah Regency, causing outbreak; attributed to dog meat trade and reduction in rabies vaccination		Mass dog vaccination	Outbreak		Wirawan et al., 2023 (56); Ward and Wera, 2025 (18)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										coverage due to COVID-19 pandemic					
Asia	Taiwan	2023	Formosan ferret-badger	Asia	Taiwan		Natural	In-country	By land/water (presumed)	Suspected incursion into Miaoli County from southern region by crossing Daan River, which experienced near-dry conditions due to decreased rainfall and controlled water diversion	No		No local transmission	No	Lai et al., 2025 (57)
Asia	Timor-Leste	Feb-24	Dog	Asia	Indonesia			International (shared border)	By land	Incursion from West-Timor, Indonesia into Oecusse, Timor-Leste, causing outbreak leading to human deaths		Increased public awareness	Outbreak	No	Anjum 2024 (58); Ward and Wera, 2025 (18)
Europe	France	Mar-01	Dog	Africa	Morocco (via Spain)	7 weeks	Human-mediated	Intercontinental, International (shared border)	By land, air/water (presumed)	Adopted by owners during camping trip in Morocco; no land border controls observed for rabies (FMD only); delayed reporting of contact with another dog by owners, resulting in fine for withholding information	No	Contact tracing, PEP, quarantine, controlled movement of dogs, sacrifice of associated animals (dog)	No local transmission	No	Bruyere-Masson et al., 2001 (59); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Albania	Mar-01	Dog	Europe	Kosovo		Natural	International (shared border)	By land	Migrated and bit 3 humans; suspected infection by wild animal (wolf/fox) as no ORV along border	No	PEP	No local transmission	No	WHO Collaborating Centre for Rabies Surveillance and Research, 2001 (62); Blanton et al., 2007 (63); Korro et al.,

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
															2009 (64); Lika, 2010 (65)
Europe	Austria	Oct-01	Dog	Europe	Serbia		Human-mediated	International	By land (presumed)	Illegally imported and sold to new owners	No	PEP, contact tracing, quarantine of exposed animals (dogs), sacrifice of associated animals (second dog of owners)	No local transmission	No	Office International des Epizooties, 2001 (66); McElhinney et al., 2011 (67)
Europe	Germany	Nov-01	Dog	Asia	Azerbaijan	2 days	Human-mediated	Intercontinental	By air	Adopted by worker; vaccinated and given health certificate without time for rabies antibodies to develop before importation	No	PEP, contact tracing	No local transmission	No	WHO Collaborating Centre for Rabies Surveillance and Research, 2001 (68); Johnson et al., 2011(17); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Germany	2001	Dog	Asia	Nepal		Human-mediated	Intercontinental	By air	Imported	No		No local transmission	No	Cliquet and Wasniewski, 2018 (31)
Europe	Austria	Jan-02	Fox	Europe	Slovenia		Natural	International (shared border)	By land	Caused outbreak resulting in >20 rabid animals	No	ORV	Outbreak	No	Potzsch, 2014 (69); Singh et al., 2018 (70)
Europe	France	2002	Dog	Africa	Morocco (via Spain)	7 weeks	Human-mediated	Intercontinental	By land, air/water (presumed)	Illegally imported without vaccination certificate	No	PEP	No local transmission	No	WHO Collaborating Centre for Rabies Surveillance and Research, 2001 (68); Lardon, 2010 (71); Johnson et al., 2011 (17); Mailles et al., 2011 (72); Ribadeau-Dumas, 2016 (60);

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
															Alvarez et al., 2022 (61)
Europe	Switzerland	May-03	Dog	Africa	Algeria or Morocco	1 month	Human-mediated	Intercontinental		Suspected to have been illegally imported; found abandoned in Switzerland; and brought to animal shelter, later adopted	No	PEP, vaccination of exposed animals	No local transmission	No	Zanoni and Breitenmoser, 2003 (73); Johnson et al., 2011 (17); Ribadeau-Dumas, 2016 (60)
Europe	Finland	Jun-03	Horse	Europe	Estonia	1 month	Human-mediated	International	By land/water	Imported without vaccination, in accordance with local guidelines (rabies vaccination recommended but not required, contrary to OIE standards); was released to pasture with other horses; bit veterinarian during examination	No	PEP (veterinarian), vaccination of exposed animals (horses), quarantine of exposed animals (horses)	No local transmission	No	Englund and Pringle, 2003 (74); Rimhanen-Finne et al., 2009 (75); Metlin et al., 2016 (76); Zoonoosikeskus (Zoonoses Centre), 2012 (77); Dominguez et al., 2016 (78)
Europe	France	2004	Dog	Africa	Morocco (via Spain)		Human-mediated	Intercontinental, International (shared border)	By land, air/water (presumed)	Vaccinated; illegally imported	No	PEP	No local transmission	No	Servas et al., 2005 (79); Lardon, 2010 (71); Ribadeau-Dumas, 2016 (60)
Europe	France	May-04	Dog	Africa	Morocco (via Spain)		Human-mediated	Intercontinental, International (shared border)	By land, air/water (presumed)	Unvaccinated; illegally imported	No	Contact tracing, PEP	No local transmission	No	Servas et al., 2005 (79); Lardon, 2010 (71); Ribadeau-Dumas, 2016 (60)
Europe	France	Jul-04	Dog	Africa	Morocco (via Spain)	1 month	Human-mediated	Intercontinental, International (shared border)	By land, air/water (presumed)	Unvaccinated; Illegally imported via car but allowed to roam unleashed at 3	No	Culling (free-roaming dogs only) Contact tracing (hotline	No local transmission	No	Health Protection Agency, 2005 (80); Servas et al., 2005 (79); Lardon, 2010

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										summer music festivals, exposing >150 humans		established), PEP, monitoring of animals			(71); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	France	2004	Dog	Africa	Morocco (via Spain)		Human-mediated	Intercontinental, International (shared border)	By land (presumed), air/water (presumed)	Illegally imported	No		No local transmission	No	Johnson et al., 2011 (17); Alvarez et al., 2022 (61)
Europe	France	2004	Dog	Africa	Morocco (via Spain)		Human-mediated	Intercontinental, International (shared border)	By land (presumed), air/water (presumed)	Illegally imported; no pet passport	No	PEP	No local transmission	No	Johnson et al., 2011 (17); Mailles et al., 2011 (72); Alvarez et al., 2022 (61)
Europe	Germany	2004	Dog	Africa	Morocco	27 days	Human-mediated	Intercontinental	By air	Imported without complete travel requirements; died in quarantine	No	PEP, contact tracing, sacrifice of associated animals with confirmed direct exposure (cat), vaccination of associated animals (dogs)	No local transmission	No	Johnson et al., 2011 (17); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Germany	Jan-05	Fox	Europe	Germany		Natural	In-country	By land	Incursion after ORV preventive vaccination cordon	No	ORV (changed from baits to aerial, at 6-week intervals instead of every autumn)	No local transmission	No	Selhorst et al., 2012 (81); Cliquet and Wasniewski, 2018 (31)
Europe	Spain	2005	Dog	Africa	Morocco		Human-mediated	Intercontinental (shared border)	By land		No		No local transmission	No	European Food Safety Authority, 2007 (82)
Europe	France	Oct-07	Dog	Africa	Morocco (via Portugal and Spain)	15 days	Human-mediated	Intercontinental, International (shared border)	By water, land	Recently vaccinated before importation with visible injuries; had direct contact	No	Contact tracing, PEP	Secondary transmission	No	Eurosurveillance, 2008 (83); Gautret et al., 2011 (84); Johnson et al., 2011

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										with 1 dog, leading to secondary transmission to 1 dog, with France losing rabies-free status for 2 years					(17); Ribadeau-Dumas, 2016 (60); Yamada et al., 2019 (85); Alvarez et al., 2022 (61)
Europe	Belgium	Oct-07	Dog	Africa	Morocco	1 month	Human-mediated	Intercontinental	By air/water (presumed)	Adopted by owner while on vacation in Morocco; cleared for importation despite not fulfilling rabies vaccination and serology requirements; brought to dog park; resulted in Belgium losing rabies-free status for 6 months	No	Contact tracing, PEP, 6-month quarantine (visitor dogs at dog park), sacrifice of associated pets, recommended pet vaccination, enhanced surveillance, restrictions, active fox surveillance, mandatory dog leashing	No local transmission	No	European Food Safety Authority, 2007 (82); Van Gucht and Le Roux , 2008 (86); Ehnert & Galland, 2009 (87); Johnson et al., 2011 (17); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Kosovo	Oct-07	Fox	Europe	Republic of North Macedonia		Natural	International (shared border)	By land		No		No local transmission	No	Muji et al., 2012 (88)
Europe	Germany	2007	Dog	Africa	Morocco		Human-mediated	Intercontinental	By air/water (presumed)	Illegally imported	No		No local transmission	No	Ehnert & Galland, 2009 (87)
Europe	Finland	2007	Dog	Asia	India	<1 month	Human-mediated	Intercontinental	By air	Adopted from streets while attacked by stray dog and vaccinated shortly before importation	No	PEP	No local transmission	No	European Food Safety Authority, 2009; Johnson et al., 2011 (17); Zoonoosikeskus (Zoonoses Centre), 2012 (77); Väyrynen, 2020 (89)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	France	Apr-08	Dog	Africa	Gambia (via Senegal and Belgium)	2 weeks	Human-mediated	Intercontinental, International (shared border)	By air, land	Fulfilled all travel requirements except serology; newly adopted and visibly wounded but given health certificate; brought into passenger cabin of plane	No	Contact tracing, PEP, 6-month quarantine of exposed animals (cats)	No local transmission	No	Eurosurveillance, 2008 (90); Roux and Gucht, 2008 (91); European Food Safety Authority, 2010 (92); Ribadeau-Dumas, 2016 (60); Cliquet and Wasniewski, 2018 (31); Alvarez et al., 2022 (61)
Europe	United Kingdom	Apr-08	Dog	Asia	Sri Lanka	6 days	Human-mediated	Intercontinental	By air	Vaccinated; imported by rescue group along with >10 animals despite not meeting minimum age requirement for travel; symptoms manifested while in quarantine	No	Contact tracing, PEP, sacrifice of associated animals (nearby dogs at quarantine center)	No local transmission	No	Catchpole et al., 2008 (93); Fooks et al., 2008 (94); Goddard et al., 2008 (95); Health Protection Agency, 2008 (96); WHO Collaborating Centre for Rabies Surveillance & Research, 2009 (97); European Food Safety Authority, 2010 (92); Johnson et al., 2011 (17); Johnson et al., 2011 (98); Fooks and Johnson, 2015 (99); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Germany	Jun-08	Dog	Europe	Croatia	6 months	Human-mediated	International	By land	Illegally imported to animal shelter without vaccination certificate; no border control measures reported at EU border	No	Contact tracing, PEP, quarantine of exposed animals	No local transmission	No	Johnson et al., 2011 (17); Weiss et al., 2009 (100); European Food Safety Authority, 2010 (92); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	France	Oct-08	Dog	Europe	Spain	1 month	Human-mediated	International	By land	Found in car park and adopted	No	Contact tracing	No local transmission	No	Johnson et al., 2011 (17)
Europe	Italy	Oct-08	Red fox	Europe	Slovenia		Natural	International (shared border)	By land	1 fox bit 1 human, and caused rabies spread to other regions; ORV not maintained during that period	No	PEP, ORV (Emergency)	Outbreak	No	De Benedictis et al., 2008 (101); Fusaro et al., 2013 (102); Berg et al, 2015 (103); Rupprecht et al., 2020 (34); Kumar et al., 2021 (104); Lojkić et al., 2021 (105)
Europe	Italy	Nov-08	Fox	Europe	Italy		Natural	In-country	By land	Continued incursion into Belluno, Venzone Province following incursion from Slovenia; caused rabies spread to other regions	No	PEP, ORV (Emergency)	Outbreak	No	Fusaro et al., 2013 (102); Berg et al, 2015 (103); Rupprecht et al., 2020 (34); Kumar et al., 2021 (104); Lojkić et al., 2021 (105)
Europe	France	2008	Dog	Europe	Spain	18 days	Human-mediated	International (shared border)	By land, air/water (presumed)	Found on highway and recently adopted; ilegally imported without vaccination	Yes	PEP	No local transmission	No	Mailles et al., 2011 (72); Alvarez et al., 2022 (61)
Europe	Germany	2009	Dog	Europe	Croatia		Human-mediated	International	By land	Imported; no border control measures reported	No		No local transmission	No	Tietjen et al., 2011 (106)
Europe	France	2009	Dog	Asia	Afghanistan		Human-mediated	Intercontinental	By air	Imported	No		No local transmission	No	WHO Collaborating Centre for Rabies Surveillance & Research, 2009 (107); Johnson et al., 2011 (17)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	Germany	Feb-10	Dog	Europe	Bosnia and Herzegovina (via Slovenia)	22 days	Human-mediated	International (shared border)	By land	Newly bought before importation with incomplete travel documents (no record of rabies vaccination) but allowed through border control; history of biting by aggressive dog	No	Contact tracing, PEP, destroyed associated animals (unvaccinated cats)	No local transmission	No	Eismann et al., 2011 (108); Johnson et al., 2011 (17); European Food Safety Authority and European Centre for Disease Prevention and Control, 2012 (109); Ribadeau-Dumas, 2016 (60); Alvarez et al., 2022 (61)
Europe	Poland	2010	Fox	Europe	Belarus or Ukraine		Natural	International (shared border)	By land	Incursion into Rzeszow	No	ORV	No local transmission	No	Smreczak et al., 2012 (110)
Europe	Poland	2010	Fox	Europe	Belarus or Ukraine		Natural	International (shared border)	By land	Incursion into Lublin	No	ORV	No local transmission	No	Smreczak et al., 2012 (110)
Europe	Poland	2010	Fox	Europe	Russia or Ukraine		Natural	International (shared border)	By land	Incursion into Malopolskie region; caused outbreak, resulting from weak surveillance (minimal sample collection)	No	ORV (maintained)	Outbreak	No	Berg et al, 2015 (103); Ortiz et al., 2018 (111)
Europe	France	Jul-11	Dog	Africa	Morocco (via Spain)	4 days	Human-mediated	Intercontinental, International (shared border)	By water, land	Newly adopted; illegally imported without fulfilling all travel requirements (underage, unvaccinated, not microchipped, no travel certificate)	No	Contact tracing, PEP, (booster) vaccination of exposed animals (dogs), sacrifice of associated animals (cats)	No local transmission	No	Mailles et al., 2011 (72); Roberts and Lopez, 2011 (112); European Food Safety Authority and European Centre for Disease Prevention and Control, 2013 (113); Alvarez et al., 2022 (61)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	Norway	2011	Arctic fox	Europe	Norway		Natural	In-country	By water (frozen)	Incursion from Svalbard mainland into Hopen Island by crossing frozen sea ice during winter; fought 4 dogs and was killed	No	(Booster) vaccination of exposed animals (dogs)	No local transmission	No	Roberts et al., 2011 (114)
Europe	Croatia	2011	Cow	Europe	Romania	8 months	Human-mediated	International	By land	One of 1,000 unvaccinated cows imported with valid documents (health certificates) but no rabies vaccination required; 6 bulls exhibited symptoms	No	Sacrifice of associated animals (bulls in same box)	No local transmission	No	Lojkić et al., 2013 (115)
Europe	Norway	2011	Arctic fox	Europe	Russia		Natural	International	By water (frozen)	Incursion into Svalbard by crossing frozen sea ice	No		No local transmission	No	Simon et al., 2021 (116); Hueffer, 2022 (117)
Europe	Poland	2011	Fox	Europe	Russia		Natural	International	By land		No	ORV (intensified)	No local transmission	No	Berg et al, 2015 (103)
Europe	Netherlands	Feb-12	Dog	Africa	Morocco (via Spain)	1 day	Human-mediated	Intercontinental, international	By air, land/water (presumed)	Purchased in parking lot and illegally imported without vaccination; examined at customs but pet passport and vaccination status not checked; stayed in passenger cabin of plane	No	Contact tracing, PEP, (booster) vaccination of exposed animals (dogs), sacrifice of associated animals (cats) due to lack of available venue for quarantine	No local transmission	No	Roberts and Lopez, 2012 (118); Van Rijckevorse et al., 2012 (119); Ribadeau-Dumas, 2016 (60); Veda et al., 2021 (120); Alvarez et al., 2022 (61)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	Greece	Oct-12	Red fox	Europe	Republic of North Macedonia		Natural	International (shared border)	By land	Seen exhibiting symptoms by villagers; led to secondary transmission to 6 foxes and 2 dogs; spread attributed to neglect of ORV and lack of mountains	No	ORV, improved passive surveillance, mandatory pet vaccination, increased public awareness	Secondary transmission	Yes	Tsiodras et al., 2013 (121); Tsiodras et al., 2014 (122); Giannakopoulos et al., 2016 (123); Rupprecht et al., 2020 (34); Lojkić et al., 2021 (105)
Europe	Spain	Jun-13	Dog	Africa	Morocco	2 months	Human-mediated	Intercontinental	By land, air/water (presumed)	Attempted importation to Morocco 4 months prior but denied due to lack of serology test; illegally imported through Ceuta immediately after vaccination; later escaped, attacked 5 humans and died, resulting in loss of rabies-free status in Spain for 6 months; owners fined for not disclosing all information	No	Contact tracing, PEP, mandatory pet vaccination (animals in Castilla-La Mancha restriction area), serology of all exposed dogs, sacrifice of associated animals (dog with insufficient antibodies according to serology), restricted movement of pets	No local transmission	No	Suarez-Rodriguez at al., 2013 (124); Pérez de Diego et al., 2015 (125); Ribadeau-Dumas, 2016 (60); Veda et al., 2021 (120); Alvarez et al., 2022 (61)
Europe	France	Oct-13	Cat	Africa	Morocco	11 days	Human-mediated	Intercontinental	By air	Imported with certificate of good health despite not meeting qualifications for importation (no vaccination); escaped home and was adopted twice by different owners	No	Contact tracing, PEP, monitoring of vaccinated animals, sacrifice of associated animals (non-vaccinated), restricted movement of pets, increased surveillance	No local transmission	No	Veterinary Record, 2013 (126); Ribadeau-Dumas, 2016 (60)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	Slovakia	2013	Dog, fox, marten	Europe	Poland		Natural	International (shared border)	By land	Incursion despite ORV	No	ORV (continued)	No local transmission	No	Ondrejková et al., 2020 (127)
Europe	France	May-15	Dog	Europe	Algeria	9 days	Human-mediated	International	By air/land (presumed)	Illegally imported without fulfilling travel requirements (no vaccination, no identification, underage); brought to Algeria 4 months later by owners despite surveillance orders, escaped for a period; manifested symptoms upon return and had direct contact with 9 humans and 1 dog	No	Contact tracing, PEP, sacrifice of associated animals (unvaccinated), monitoring of vaccinated animals (dogs) for 6 months, catching of stray animals, restricted movement of pets, increased surveillance	No local transmission	No	European Food Safety Authority and European Centre for Disease Prevention and Control, 2015 (128); Veterinary Record, (2015) (129); Veda et al., 2021 (120); Alvarez et al., 2022 (61)
Europe	Slovakia	2015	Fox	Europe	Poland		Natural	International (shared border)	By land		No	ORV	No local transmission	No	Ondrejková et al., 2020 (127)
Europe	United Kingdom	Jan-20	Sable	Europe	Russia (via Italy)		Human-mediated	International	By air (presumed)	Imported	No		No local transmission	No	European Food Safety Authority and European Centre for Disease Prevention and Control, 2021 (130)
Europe	France	Feb-20	Dog	Africa	Morocco		Human-mediated	Intercontinental	By air/water (presumed)	Illegally imported (no vaccination)	No	PEP	No local transmission	No	Bacigalupo et al., 2022 (131); European Food Safety Authority and European Centre for Disease Prevention and Control, 2021 (130); Veda et al., 2021 (120); Alvarez et al., 2022 (61)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
Europe	Poland	Jan-21	Fox	Europe	Belarus or Ukraine		Natural	International (shared border)	By land	Observed to be 'sick', killed by dog; spread facilitated by limitation of ORV to borders only	No	Increased public awareness, increased rabies surveillance, hunting ban, cancellation of pet-centric events, mandatory leashing of dogs, emergency ORV	No local transmission	No	Smreczak et al., 2023 (132)
Europe	Germany	Jul-21	Dog	Asia	Republic of Türkiye (via Bulgaria)	11 days	Human-mediated	Intercontinental, international	By air (presumed)	Illegally imported (no vaccination)	No	Contact tracing, PEP	No local transmission	No	Alvarez et al., 2022 (61); European Food Safety Authority and European Centre for Disease Prevention and Control, 2022 (133)
Europe	Slovakia	Sep-22	Badger	Europe	Poland or Ukraine		Natural	International (shared border)	By land	Suspected incursion from Poland/Ukraine; spread attributed to war in Ukraine and COVID-19 pandemic challenging rabies control measures	Yes	Enhanced surveillance	No local transmission	Yes	Robardet et al., 2023 (134)
Europe	France	Oct-22	Dog	Europe	Morocco		Human-mediated	Intercontinental	By air/water (presumed)	Suspected to be illegally imported to shelter; bit several humans	No	Contact tracing, PEP, monitoring of shelter	No local transmission	No	Bacigalupo et al., 2022 (131)
Europe	Hungary	2022	Fox	Europe	Ukraine		Natural	International (shared border)	By land	Suspected incursion from Ukraine into Hungary; spread attributed to war in Ukraine and COVID-19 pandemic	Yes	Enhanced surveillance	No local transmission	Yes	Robardet et al., 2023 (134); Marko et al., 2025 (135)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										challenging rabies control measures					
Europe	France	2024	Dog	Africa	Morocco		Human-mediated	Intercontinental		Imported from Morocco to France	No		No local transmission	No	European Food Safety Authority 2025 (136)
North America	United States of America	2001	Raccoon	North America	United States of America		Natural	In-country	By land	Incursion from Pennsylvania into Ohio despite immune corridor from ORV; attributed to weakened surveillance	No	ORV (expanded)	No local transmission	No	Krebs et al., 2002 (137)
North America	Canada	Dec-02	Arctic fox	Europe	Greenland		Natural	Intercontinental	By water (frozen)	Incursion by crossing through packed ice during winter	Yes		No local transmission	No	Nadin-Davis et al., 2008 (138)
North America	Canada	2002	Fox	North America	Canada		Natural	In-country	By water (frozen)	Incursion from mainland to Cartwright, Labrador by crossing through packed ice; overlap with red fox populations facilitated spread in arctic foxes	No		No local transmission	No	Nadin-Davis et al., 2008 (138)
North America	United States of America	2003	Dog	North America	United States of America		Human-mediated	In-country	By air	Imported	No	PEP	No local transmission	No	Lankau et al., 2014 (139)
North America	United States of America	Mar-04	Dog	North America	Mexico		Natural	International (shared border)	By land		No	Mass dog vaccination, ORV, monitoring	No local transmission	No	Blanton et al., 2007 (63)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
North America	United States of America	Jun-04	Dog	Asia	Thailand	2 days	Human-mediated	Intercontinental	By air	newly adopted and imported without vaccination; given health certificate despite history of respiratory illness; stayed in passenger cabin	No	Contact tracing, PEP	No local transmission	No	Castrodale et al., 2008 (140); McQuiston, 2008 (141); Ehnert & Galland, 2009 (87); Lankau et al., 2014 (139)
North America	United States of America	Jul-04	Raccoon	North America	United States of America		Natural	In-country	By land	Suspected incursion from mid-Atlantic state to Ohio; detected despite weakened surveillance, limited testing	Yes	ORV, enhanced surveillance	No local transmission	No	Russell et al., 2005 (142); Henderson et al., 2008 (143); Slate et al., 2008 (144)
North America	United States of America	2004	Raccoon	North America	United States of America		Natural	In-country	By land	Incursion into Cape Cod from nearby areas; crossed vaccine barrier and led to outbreak in raccoons with transmission to other species (skunks, domestic animals, etc.)	No	ORV, trap-vaccination-release of raccoons and skunks	Outbreak	Yes	Wang et al., 2009 (145)
North America	United States of America	Feb-06	Red fox	North America	United States of America		Natural	In-country	By land	Incursion from mid-Atlantic state to Tennessee; found during enhanced surveillance in non-ORV area; suspected infection by raccoon; ORV not recommended, considered 'waste of resources'	Yes	Enhanced surveillance (increased sampling)	No local transmission	No	Slate et al., 2008 (144)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
North America	Canada	Jun-06	Raccoon	North America	United States of America		Natural	International (shared border)	By land			ORV	No local transmission	No	Shwiff et al., 2013 (146); Stevenson et al., 2016 (147); Trewby et al., 2017 (148); Nadin-Davis, 2018 (149); Nadin-Davis et al., 2020 (150)
North America	United States of America	Mar-07	Dog	Asia	India	2 days	Human-mediated	Intercontinental, in-country	By air	Newly adopted before importation by volunteer veterinarian; no vaccination; given to 2nd veterinarian; despite symptoms and biting history (veterinarian owner, dog), health certificate obtained from 3rd veterinarian for travel	No	PEP, contact tracing, sacrifice of associated pets (dog)	No local transmission	No	Blanton et al., 2008 (151); Castrodale et al., 2008 (140); Ehnert & Galland, 2009 (87); Lankau et al., 2014 (139)
North America	United States of America	2008	Dog	Asia	Iraq	3 days	Human-mediated	Intercontinental	By air	Adopted by soldier and kept on military base for 7 months before importation; transported with >20 dogs with no vaccination certificates, leading to 30-day quarantine for all animals; diagnosed in quarantine centre	No	Contact tracing, PEP, vaccination of associated animals (dogs, cat); 6-month quarantine of exposed animals (dogs, cat)	No local transmission	No	Mangieri et al., 2008 (152); Lankau et al., 2014 (139); Hercules et al., 2018 (153)
North America	United States of America	Jan-10	Cow	North America	Mexico	5 months	Human-mediated	International (shared border)	By land	Imported without documentation; presumed unvaccinated	Yes		No local transmission	No	Blanton et al., 2011 (154); Pieracci et al., 2020 (155)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										suspected infection by vampire bat					
North America	United States of America	Jan-13	Raccoon	North America	United States of America		Natural	In-country	By land	Incursion from mainland Georgia to Jekyll Island by crossing bridge; led to outbreak	No	Outbreak ended in 2017	Outbreak	No	Ortiz et al., 2018 (111)
North America	Canada	Jul-13	Dog	North America	Canada	1 month	Human-mediated	In-country	By air	Found in area with ongoing outbreak, scavenging for food with other dogs (mother and newborn litter); not vaccinated before importation due to no vaccination requirement for in-country travel; suspected infection by arctic fox	Yes	PEP, quarantine of exposed animals (mother and litter dogs), sacrifice of associated animals (dog negative for rabies but euthanized at request of owner)	Outbreak	Yes	Curry et al., 2016 (156)
North America	Canada	Dec-14	Dog	North America	Canada	0 days	Human-mediated	In-country	By air, land	Found in construction work camp; not vaccinated before importation due to no vaccination requirement for in-country travel; manifested symptoms during travel and bit owner; suspected infection by arctic fox	Yes	PEP, vaccination of exposed animal (dog), 45-day quarantine of exposed animal (dog)	No local transmission	No	Curry et al., 2016 (156)
North America	United States of America	May-15	Dog	Africa	Egypt	4 days	Human-mediated	Intercontinental	By air	Found on street and imported by animal rescue organization along with >30 pets	No	Contact tracing, PEP, vaccination of exposed animals (puppy, booster for	No local transmission	No	Sinclair et al., 2015 (157); Pieracci et al.,

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										(dogs, cats) despite fracture injury; transported in same crate with own puppy; falsified vaccination certificate		vaccinated dogs), quarantine of exposed dogs (puppy)			2020 (155); Latzer et al., 2022 (158)
North America	Canada	Dec-15	Raccoon	North America	United States of America		Human-mediated	International (shared border)	By land/water (presumed)	Suspected long-distance incursion by stowing away on truck or ship; picked up by animal control along with 2 dogs; escaped cage and fought with dog; led to outbreak (>400 animals)	No	PEP, mass vaccination, ORV, enhanced surveillance (increased testing of sick animals), increased public awareness	Outbreak	Yes	Stevenson et al., 2016 (147); Trewby et al., 2017 (148); Lobo et al., 2018 (159); Gilbert & Chipman, 2020 (160); Nadin-Davis et al., 2020 (150)
North America	Canada	2015	Raccoon	North America	United States of America		Natural	International (shared border)	By land		Yes		No local transmission	No	Birhane et al., 2017 (161); Trewby et al., 2017 (148); Nadin-Davis et al., 2020 (150)
North America	Canada	2015	Raccoon	North America	United States of America		Natural	International (shared border)	By land		Yes		No local transmission	No	Trewby et al., 2017 (148)
North America	United States of America	Mar-16	Raccoon	North America	United States of America		Human-mediated	In-country	By land	Suspected human-mediated incursion from Connecticut into New York	Yes	Enhanced surveillance, increased public awareness	No local transmission	No	Brunt et al., 2020 (162)
North America	United States of America	Dec-16	Cat	North America	United States of America		Human-mediated	In-country	By air/land (presumed)	Imported from Iowa to New York for vacation without vaccination; attacked 1 dog, 3 humans;	Yes		No local transmission	No	Brunt et al., 2021 (163)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										suspected infection by skunk					
North America	United States of America	Dec-16	Otter	North America	United States of America		Natural	In-country	By water	Suspected incursion from Connecticut into New York	Yes		No local transmission	No	Brunt et al., 2020 (162)
North America	United States of America	Feb-17	Cat	North America	United States of America	5 months	Human-mediated	In-country	By air/land (presumed)	Newly adopted before importation from North Carolina to Arkansas along with >10 other pets (cats, dogs) then later surrendered to Humane Society of Summit County in Ohio; vaccinated but not for rabies (according to facility guidelines, rabies vaccination only administered at time of spay/neuter); suspected infection by raccoon	Yes	PEP, vaccination of exposed animals (recommended), 6-month quarantine of exposed animals (cats)	No local transmission	No	Singh et al., 2018 (70)
North America	United States of America	Nov-17	Cat	North America	United States of America			In-country	By land (presumed)	Imported from Westchester County to Long Island without vaccination; bit veterinary staff; suspected infection by raccoon	Yes		No local transmission	No	Brunt et al., 2020 (162)
North America	United States of America	Dec-17	Dog	Africa	Egypt	1 day	Human-mediated	Intercontinental	By air	Iimported from with 3 other dogs by animal rescue organization despite visible injuries; suspected falsified	No	Contact tracing, PEP, booster vaccination of exposed animals (dogs), quarantine	No local transmission	No	Blanton et al., 2009 (164); Hercules et al., 2018 (153); Chevalier & Havas, 2019 (165);

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										rabies vaccination document; bit 1 human (flight parent) before boarding plane		of exposed animals (dogs), monitoring			Latzer et al., 2022 (158)
North America	United States of America	Jan-19	Dog	Africa	Egypt (via Canada)	1 month	Human-mediated	Intercontinental, international (shared border)	By air, land	Recently fostered before importation with >20 other dogs; bit 1 human (veterinary technician) during examination; had rabies vaccination certificate but serologic testing showed lack of vaccination in some dogs, indicating vaccination failure or falsification of vaccination certificate	No	PEP, suspension of dogs entering USA from Egypt, vaccination of exposed animals (booster) and 45-day monitoring, 6-month quarantine of exposed animals (unvaccinated dogs), quarantine of imported animals (dogs) and serologic monitoring	No local transmission	No	Raybern et al, 2019 (166); Latzer et al., 2022 (158)
North America	United States of America	May-21	Anteater	North America	United States of America	2 months	Human-mediated	In-country	By air/land (presumed)	Imported from Virginia zoo to Tennessee Zoo; suspected infection by raccoon	Yes	Contact tracing, PEP, vaccination of exposed animal (tamandua) that was presumed unvaccinated due to missing vaccination records), quarantine of exposed animal (tamandua)	No local transmission	No	Grome et al, 2022 (167)
North America	United States of America	Jun-21	Dog	Asia	Azerbaijan	3 days	Human-mediated	Intercontinental	By air, other	Imported with >30 other animals (dogs, cat) by animal rescue organization;	No	Contact tracing, PEP, (booster) vaccination of exposed animals	No local transmission	No	Williams & Pieracci, 2021 (168); Whitehill et al., 2022 (169)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										serologic testing showed insufficient levels despite confirmed vaccination, indicating vaccination failure due to underdosing		(dogs, cat), serologic monitoring and 45-day quarantine of imported animals (dogs), suspension of dog importations from DMRVV high-risk countries			
North America	Canada	Jul-21	Dog	Asia	Iran (via Germany)	11 days	Human-mediated	Intercontinental	By air	Newly adopted and imported by animal rescue organization; presumed vaccinated but was not revaccinated upon arrival despite vaccination policy for young dogs	No	Contact tracing, PEP, (booster) vaccination of exposed animals (dog), 3-month quarantine of exposed animal (dog) due to delayed identification	No local transmission	No	Rebellato et al., 2022 (170)
North America	Canada	Jun-21	Dog	Asia	Iran (via Germany)	11 days	Human-mediated	Intercontinental	By air	Newly adopted and imported by animal rescue organization as personally owned pet; met importation requirements as no waiting period required between vaccination and travel for personal pets	No	Contact tracing, PEP, (booster) vaccination of exposed animals (dog), 3-month quarantine of exposed animal (dog) due to delayed identification	No local transmission	No	Belanger et al., 2025 (171); Di Salvo et al., 2023 (172)
North America	United States of America	Mar-22	Coyote	North America	United States of America		Natural	In-country	By land	Incursion from Monongalia County into Lewis County, West Virginia; suspected infection by raccoon indicated by phylogenetic analysis; ORV not	Yes		No local transmission	No	Hopken et al., 2024 (173)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										recommended due to low chance of onward local transmission					
North America	United States of America	2023	Skunk	North America	United States of America		Human-mediated	In-country	By land	Purchased from breeder in Attica and brought home to northern Michigan	No		No local transmission	No	Boutelle et al., 2025 (174)
South America	Brazil	2006	Dog	South America	Bolivia		Natural	International (shared border)	By land	Attributed to high numbers of free-roaming dogs	No	Mass dog vaccination	No local transmission	No	Galhardo et al., 2019 (175)
South America	Peru	2015	Dog	South America	Peru		Natural	In-country	By land	Incursion from Puno into Arequipa; attributed to low vaccination rate among free-roaming dogs, commonly found foraging in water channels throughout city that collect garbage for foraging during dry season; led to outbreak	No	Mass dog vaccination, culling (stray dogs)	Outbreak	No	Castillo-Neyra et al., 2017 (176); Castillo-Neyra et al., 2017 (177); Raynor et al., 2020 (178)
South America	Brazil	2015	Dog	South America	Bolivia		Natural	International (shared border)	By land	Further spread led to outbreak in Mato Grosso do Sul	No	Outbreak ended in 2015	Outbreak	No	Benavides et al., 2019 (179)
South America	Peru	Feb-21	Dog	South America	Peru		Human-mediated	In-country	By land	Imported from Puno to El Pedregal, Majes District, Caylloma Province as indicated by phylogenetic analysis; bit local dog that resulted in	Yes	Contact tracing, PEP, mass dog vaccination, increased public awareness;	Outbreak	No	Salazar et al., 2025 (180)

Continent	Incursion location	Incursion date	Species	Origin continent	Location origin	Time between incursion & symptoms	Type of incursion	Borders crossed	Mode of travel	Incursion details	Phylogenetic analysis	Public health response	Local transmission	Unusual cross-species transmission	Reference(s)
										outbreak; attributed to weak border controls, difficulty vaccinating free-roaming dogs and lapse in annual mass dog vaccination due to COVID-19 pandemic		outbreak controlled in 2021			
South America	Peru	Sep-22	Dog	South America	Peru		Human-mediated	In-country	By land	Imported from Arequipa into El Pedregal, resulting in outbreak	Yes	Contact tracing, PEP, mass dog vaccination, increased public awareness; outbreak controlled in 2022	Outbreak	No	Salazar et al., 2025 (180)

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